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Order–disorder transition of β-brass (CuZn) by Metropolis Monte Carlo on a rigid lattice

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Abstract

We wrote a Metropolis Monte Carlo code that exchanges Cu and Zn atoms on a fixed body-centred cubic lattice of 432 sites, with a pair energy model limited to first and second neighbours. The code reproduces the exact values of the perfect B2 structure (cohesive energy 2.484 eV/atom) and of a random alloy. Scanning the temperature from 200 K to 1600 K, we follow the energy, the heat capacity, the long-range order parameter η, its susceptibility and the Cowley short-range order parameters. The three estimates of the transition temperature all give $T_c \approx 720$ K, close to the experimental value of about 740 K. Short-range order survives above T_c.

1 Introduction

β-brass is the CuZn alloy with about 50 % Zn. At low temperature it is ordered in the B2 (CsCl-type) structure: the atoms sit on a body-centred cubic (bcc) lattice, Cu on the cube corners and Zn on the cube centres. When the temperature rises, Cu and Zn atoms swap places more and more often. Around 740 K the distinction between the two kinds of sites disappears: the alloy becomes a disordered bcc solid solution. This is a classic order–disorder transition, first described by Bragg and Williams with a mean-field theory.

In this project the atoms do not move away from the lattice sites ("rigid lattice"): only the occupation of each site by Cu or Zn changes. The energy is a sum of pair interactions between first and second neighbours. We sample the equilibrium configurations with the Metropolis algorithm and measure how the order is lost when the temperature increases.

2 Model and method

2.1 Lattice and neighbours

At low temperature CuZn has the ordered B2 structure (CsCl type), made of two interpenetrating simple cubic sublattices α and β. The cubic cell of parameter a contains two sites (Figure 1): the α site at the corner $(0, 0, 0)$, occupied by Cu in B2, and the β site at the centre $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, occupied by Zn. Whatever the species on them, the sites form a bcc lattice, which stays fixed during the simulation. The simulation box contains $6 \times 6 \times 6$ cells, that is $N = 2 \times 6^3 = 432$ sites (216 Cu and 216 Zn, $c_{\mathrm{Zn}} = 1/2$), with periodic boundary conditions in the three directions. The reference configuration at $T = 0$ K is the B2 cell repeated $6 \times 6 \times 6$ times.

Each site has two neighbour shells (Table 1): 8 first neighbours at distance $\sqrt{3}a/2$, all on the other sublattice, and 6 second neighbours at distance a, on the same sublattice. The list of these 14 neighbours is built once for every site at the start of the program, applying the periodic boundary

conditions (modulo 6 on each cell coordinate). This table never changes: only the occupation of the sites does.

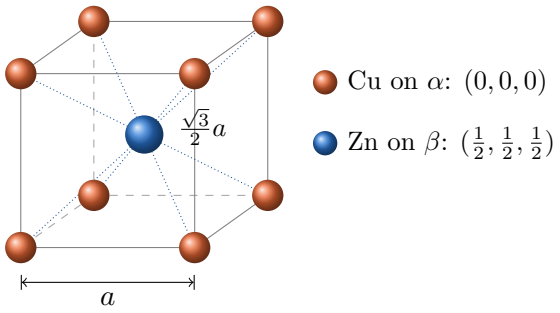


Figure 1: Cell of the B2 structure of CuZn (perfect order, $T = 0\,\mathrm{K}$). Cu atoms (orange) occupy the corners (sublattice α), Zn atoms (blue) the centre (sublattice β). The dotted lines are the 8 first-neighbour bonds of the centre atom (length $\sqrt{3}a/2$); second neighbours are along the cube edges (length a). In the disordered phase, Cu and Zn occupy both sublattices at random, and the lattice is simply bcc.

2.2 Energy

The energy of a configuration is

$$E = \sum_{\langle i,j\rangle_1} J_{\sigma_i\sigma_j} + \sum_{\langle i,j\rangle_2} I_{\sigma_i\sigma_j}, \tag{1}$$

where $\sigma_i \in \{\mathrm{Cu}, \mathrm{Zn}\}$ is the atom on site i, and the sums run over first- and second-neighbour pairs, each pair counted once. The pair energies are given in Table 1.

Table 1: Neighbour shells and pair energies of the model in eV. J acts between first neighbours, I between second neighbours. The last column is the "ordering energy" $w = X_{\mathrm{CuCu}} + X_{\mathrm{ZnZn}} - 2X_{\mathrm{CuZn}}$ of each shell ($X = J$ or I): $w > 0$ favours unlike pairs, $w < 0$ favours like pairs.

Shell	Distance	Neighbours	Cu–Cu	Zn–Zn	Cu–Zn	w
1st neighbours (J)	$\sqrt{3}a/2$	8	-0.873	-0.338	-0.621	$+0.031$
2nd neighbours (I)	a	6	0.000	0.000	$+0.005$	-0.010

The first shell favours Cu–Zn pairs ($w_1 > 0$) and the second shell favours pairs of the same species ($w_2 < 0$). In the B2 structure the 8 first neighbours of every atom are of the other species and the 6 second neighbours of the same species, so both shells push towards B2 order.

2.3 Monte Carlo simulation

We sample the Boltzmann distribution $P \propto e^{-E/k_BT}$ with the Metropolis algorithm [1], whose main lines are given in Algorithm 1. Starting from the perfect B2 structure, each step draws a random atom i and one of its 8 first neighbours j, computes the energy E_{after} after exchanging them, and accepts the exchange with probability $\min\left(1, e^{-(E_{\mathrm{after}}-E)/k_BT}\right)$; otherwise the configuration is kept. One *sweep* is $N = 432$ steps, so that every atom is drawn once on average (this is purely our choice to use this number for one sweep, it has no theorical meaning. We explain why in part 2.4). The first n_{eq} steps are discarded (Section 3), then the quantities of Section 2.4 are measured every m steps.

2.4 Measured quantities

After equilibration, we record once per sweep the energy E, the long-range order parameter and the two Cowley parameters. We chose to record those values every sweep and not every step, as one

Algorithm 1 Metropolis. The black lines are common to both versions of the run: the blue lines are the measurement run (quantities recorded every m steps after equilibration), the orange line is the equilibration study (energy recorded after every step).

Require: T, n_{eq}, n_{after}, m (`measure_every`)
Initialize the system at $0K$
Compute the initial total energy E
Initialize empty lists *energies*, η, α_1, α_2
for $k = 0$ to $n_{\text{eq}} + n_{\text{after}} - 1$ **do**
    Randomly select an atom i
    Randomly select one of its 8 neighbours j
    Compute the energy after the swap E_{after}
    Calculate the acceptance probability: $P \leftarrow \min\left(1, e^{-(E_{\text{after}} - E)/(k_B T)}\right)$
    if the swap is accepted with probability P **then**
        Perform the swap between i and j
        $E \leftarrow E_{\text{after}}$
    end if
    Append E to *energies*
    if $k \geq n_{\text{eq}}$ **and** $(k - n_{\text{eq}}) \bmod m = 0$ **then**
        Compute α_1, α_2 (Cowley) and $|\eta|$
        Append E to *energies*
        Append $|\eta|, \alpha_1, \alpha_2$ to their respective lists
    end if
end for
return *energies*, η, α_1, α_2

step might do nothing and it is better to let the system evolve a little bit before sampling any value. This is purely our choice.

$$\eta = \frac{N_{\text{Cu}}^{(\alpha)} - N_{\text{Cu}}^{(\beta)}}{N_{\text{Cu}}^{(\alpha)} + N_{\text{Cu}}^{(\beta)}}, \qquad \alpha_n = 1 - \frac{p_n^{\text{Zn|Cu}}}{c_{\text{Zn}}} \quad (n = 1, 2), \tag{2}$$

Here $N_{\text{Cu}}^{(\alpha)}$ is the number of Cu atoms on sublattice α, $p_n^{\text{Zn|Cu}}$ the probability that a neighbour of shell n of a Cu atom is a Zn atom, and $c_{\text{Zn}} = 1/2$. The perfect B2 order gives $\eta = 1$, $\alpha_1 = -1$ and $\alpha_2 = +1$; a random alloy gives $\eta = \alpha_1 = \alpha_2 = 0$. In a finite box the ordered state can also form with Cu on β ($\eta < 0$), and near T_c the system goes back and forth between the two. We therefore average $|\eta|$, which measures the amount of order whatever its sign.

From the series of measurements we compute the means $\langle E \rangle$, $\langle |\eta| \rangle$, $\langle \alpha_n \rangle$, and two response functions from the fluctuations:

$$C_v = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}, \qquad \chi = \frac{N\left(\langle \eta^2 \rangle - \langle |\eta| \rangle^2\right)}{k_B T}. \tag{3}$$

C_v is the heat capacity of the whole box (in eV/K) and χ the susceptibility of the order parameter (in eV^{-1}).

3 Equilibration

Every run starts from the perfect B2 structure, which is not representative of the equilibrium. The first steps must be discarded until the system has "forgotten" this initial state. To choose this number of steps, we record the energy after every step during 200 000 steps at each temperature of the scan (Figure 2). The equilibrium energy E_{eq} and the size σ of its fluctuations are the mean and standard deviation over the second half of the run; equilibrium is reached at the first step where $E \geq E_{\text{eq}} - \sigma$.

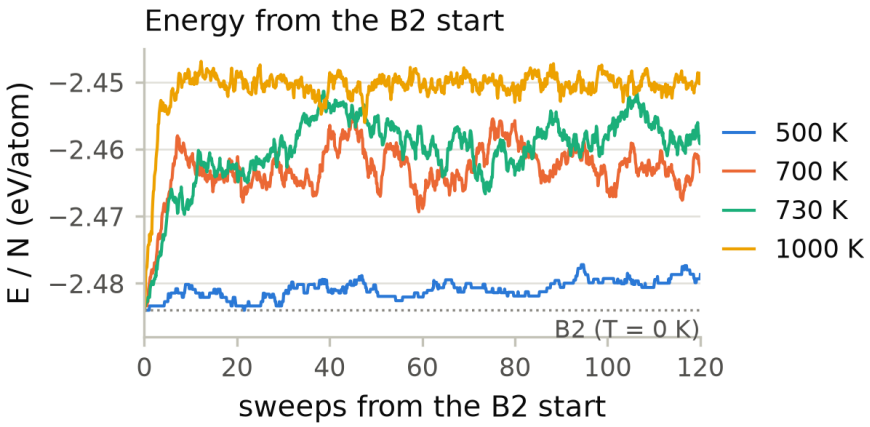


Figure 2: Equilibration from the perfect B2 structure: energy per atom during the first 120 sweeps at four temperatures (one value every 50 steps). The dotted line is the B2 energy.

At low temperature (200–300 K) the B2 structure is already the equilibrium state and almost no exchange is accepted. At high temperature, disorder sets in within a few thousand steps. The slowest relaxations are found below and near the transition, with at most 11 806 steps (27 sweeps) at 730 K. This estimate is itself noisy, because the energy fluctuates strongly near T_c. We therefore discard $N_{eq} = 50\,000$ steps at every temperature, more than four times the slowest relaxation.

4 Physical quantities and transition temperature

At each temperature tested, we ran the Metropolis algorithm from the B2 structure until equilibrium (N_{eq} steps, Section 3). Once equilibrium was reached, we continued the run for 4000 sweeps and calculated the physical quantities of Section 2.4 from the recorded series. All the values are listed in Table 3.

4.1 Evolution with temperature

Figure 3 shows what the configurations look like. At 300 K the layer is a perfect checkerboard of Cu and Zn. Near T_c the order is only partial, and at 1200 K the atoms are spread at random over the two sublattices.

Energy and heat capacity. The mean energy (Figure 4, left) starts from the B2 value and rises steeply between 600 and 800 K, where most of the disorder appears. The heat capacity (Figure 4, right) has a clear peak at 720 K; its value at 700 K is almost the same, so the top of the peak is between 700 and 720 K.

Long-range order. $\langle|\eta|\rangle$ (Figure 5, left) decreases slowly up to about 600 K, then drops fast between 650 and 800 K, and finally approaches the value of a random lattice, about 0.04. This value is not zero because, with only 216 Cu atoms, chance alone puts a few more of them on one sublattice. The susceptibility (Figure 5, right) has a sharp peak at 720 K, almost equal to its value at 730 K.

Figure 6 shows the distribution of $|\eta|$. Below T_c it is narrow around a large value; above T_c it is narrow close to zero. Near T_c it is very broad: the system wanders between ordered and disordered states.

Short-range order. Figure 7 shows the Cowley parameters. Below T_c the Cowley parameters follow the curves of η: the long-range order explains almost all the local order. Above T_c, however,

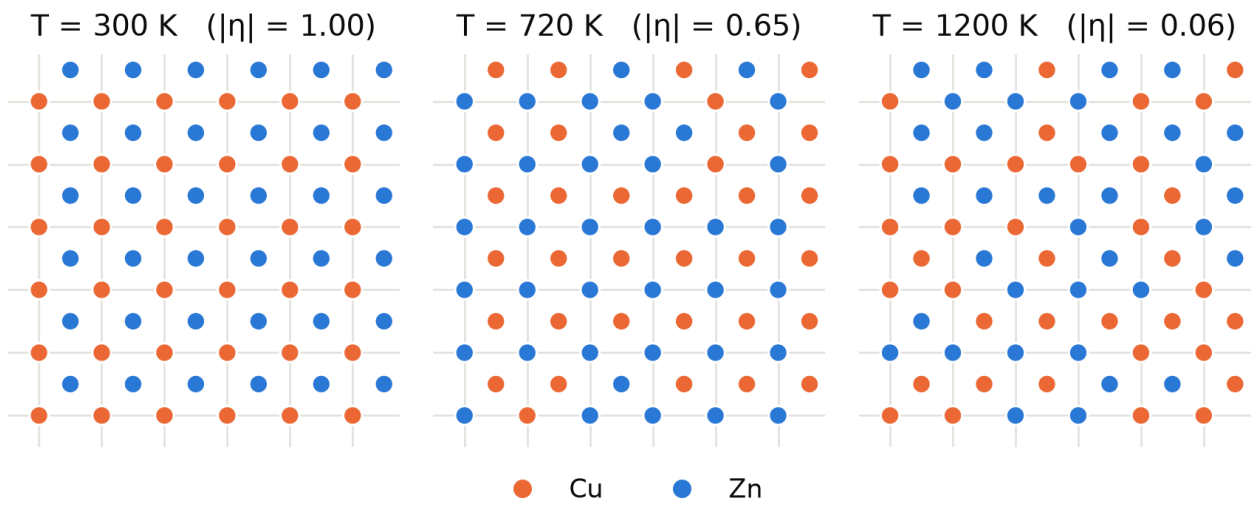


Figure 3: Final configuration of the runs at three temperatures, seen from above: one layer of the box, with the α sites of height $z = 0$ at the grid nodes and the β sites of height $z = a/2$ at the centres of the squares (6×6 cells, 72 of the 432 atoms). $|\eta|$ is computed on the whole box.

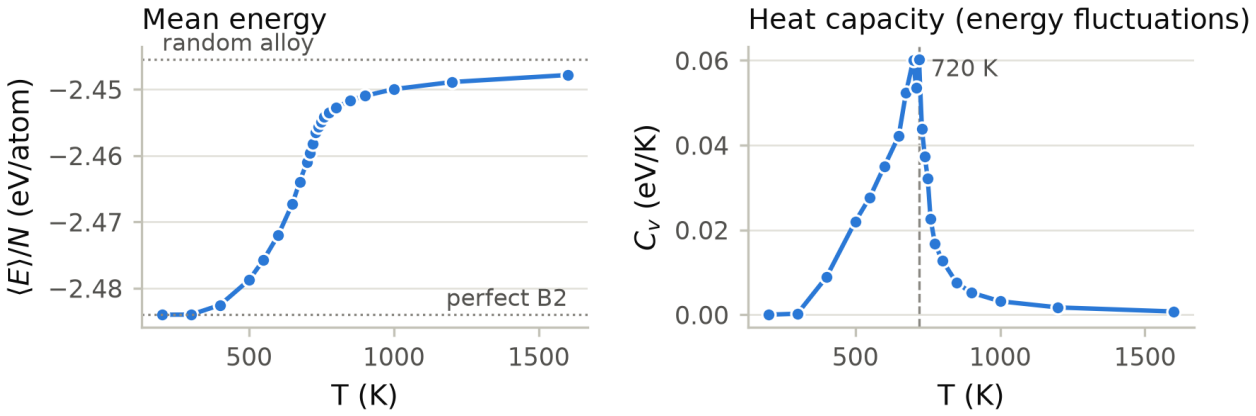


Figure 4: Left: mean energy per atom versus temperature. Right: heat capacity C_v of the box (432 atoms), from the energy fluctuations (Equation (3)); the dashed line marks its maximum.

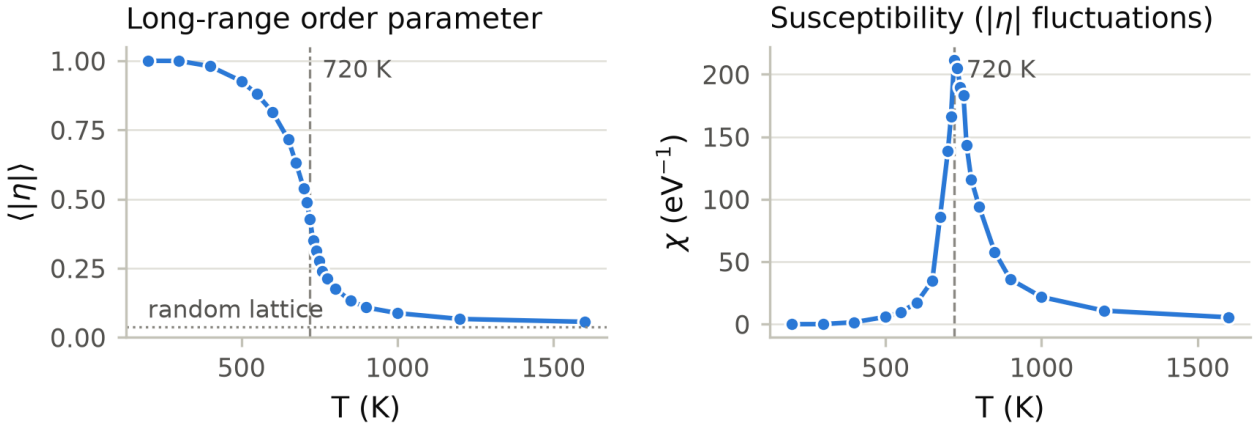


Figure 5: Left: mean absolute long-range order parameter; the dotted line is the value for random lattices, the dashed line marks the steepest drop. Right: susceptibility of the order parameter from the fluctuations of $|\eta|$ (Equation (3)); the dashed line marks its maximum.

$\langle \eta^2 \rangle$ is almost zero while α_1 stays clearly negative: at 800 K, $\langle \alpha_1 \rangle = -0.20$ whereas $-\langle \eta^2 \rangle = -0.05$, and at 1600 K $\langle \alpha_1 \rangle$ is still -0.07. Cu atoms keep more Zn first neighbours than in a random alloy: this *short-range order* survives the loss of long-range order and fades slowly with temperature, as

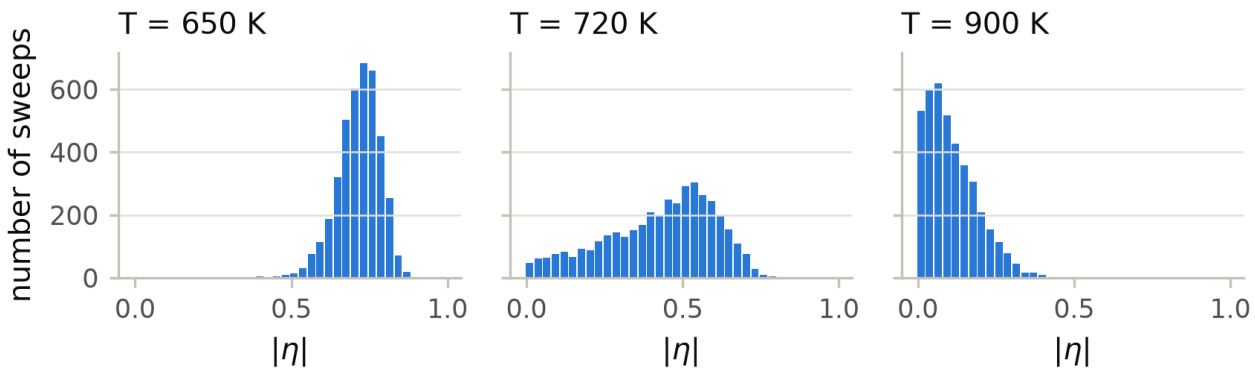


Figure 6: Histograms of the 4000 values of $|\eta|$ measured (one per sweep) below, near and above the transition. In our box $|\eta|$ can only take the values $2k/216$; each bin contains three of them (width $6/216 \approx 0.028$).

observed experimentally in ordering alloys.

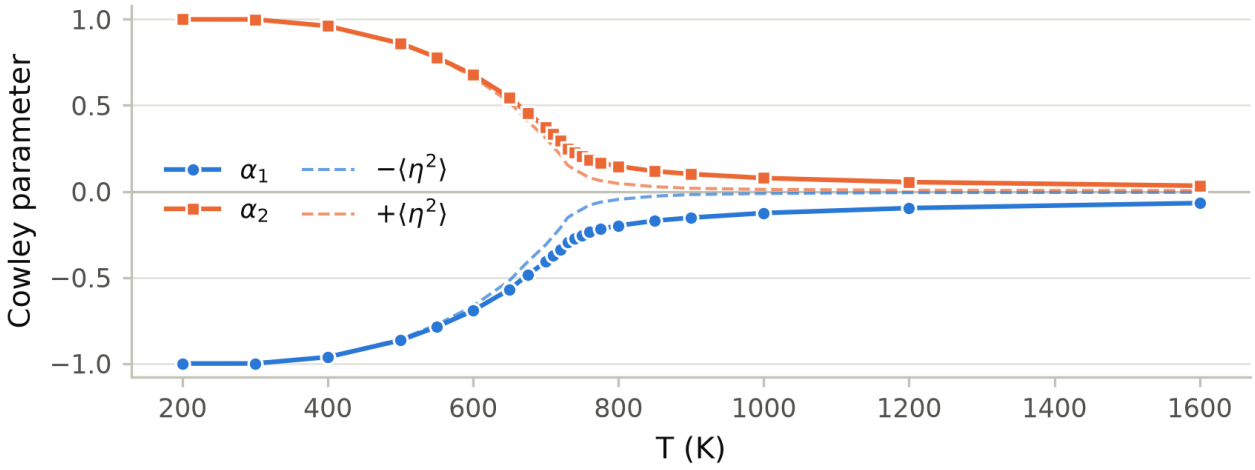


Figure 7: Cowley short-range order parameters for the first (α_1, circles) and second (α_2, squares) neighbour shells. The dashed lines $\mp\langle\eta^2\rangle$ are the values expected if the long-range order alone were present. The difference between symbols and dashed lines above 750 K is the short-range order.

4.2 Transition temperature

The energy, the order parameter and the two response functions all change abruptly around the same temperature: the alloy goes through a phase transition. Each quantity shows it in its own way:

- *Order parameter.* η distinguishes the two sublattices: it is non-zero in the ordered phase and zero in the disordered phase, where α and β become equivalent. Its fast drop is the loss of long-range order, which defines the transition.
- *Heat capacity.* $C_v = \mathrm{d}\langle E\rangle/\mathrm{d}T$ measures how fast the energy rises with temperature.
- *Susceptibility.* χ measures the fluctuations of η. Its peak shows that the order is the least stable at the transition, where the system wanders between ordered and disordered states (Figure 6).

Table 2 gives the three estimates. They fall on the same grid temperature, 720 K. The grid step is 10 K there, but the peaks are flat at the top, between 700 and 730 K (Table 3): we retain $T_c \approx 720$ K.

Table 2: Transition temperature. The Monte Carlo values are read on the temperature measured (step 10 K between 700 and 760 K).

Method	T_c (K)
Maximum of the heat capacity C_v	720
Maximum of the susceptibility χ	720
Steepest drop of $\langle \lvert\eta\rvert \rangle$	720
Experiment (β-brass) [2]	≈ 740

5 Discussion

Our estimate of T_c is close to the experimental value [2], which is quite good for a model with only six parameters and no adjustment.

Order of the transition. In a first-order transition the order parameter and the energy jump at T_c (latent heat), and the ordered and disordered phases coexist. In a second-order transition the order parameter goes to zero continuously, while the heat capacity and the susceptibility diverge. Our results look continuous: $\langle \lvert\eta\rvert \rangle$ decreases regularly through T_c (0.54 at 700 K, 0.43 at 720 K, 0.31 at 740 K, Table 3), the energy shows no visible jump, and at 720 K the histogram of $\lvert\eta\rvert$ is a single broad hump, not two separated peaks (Figure 6). This favours a second-order transition, which is the usual description of β-brass [3]. Our box of 432 atoms is too small to be sure of this. In a finite box there is no true singularity: a jump of η is rounded into a steep but continuous drop, and a divergence is cut into a finite peak (here $C_v^{\max} \approx 0.06\,\text{eV/K}$ and $\chi^{\max} \approx 210\,\text{eV}^{-1}$), because near T_c the correlation length becomes larger than the box.

6 Conclusion

We wrote and tested a Metropolis Monte Carlo code for the order–disorder transition of CuZn on a rigid bcc lattice. The tests on the perfect B2 structure give the expected values. The cohesive energy of the B2 phase is 2.484 eV/atom. Starting from B2, the system needs at most about 12 000 steps to reach equilibrium, so we discarded 50 000 steps before measuring during 4000 sweeps. The energy, the heat capacity, the long-range order parameter and its susceptibility all show the transition at $T_c \approx 720$ K, close to the experimental value (about 740 K). Above the transition, Cu and Zn still prefer to be first neighbours: short-range order survives the loss of long-range order. The main limits of this study are the small size of the box and the simple pair model; larger boxes and a finite-size analysis would be the next step.

References

[1] M. Hayoun and H. Dammak. *Simulation numérique des ensembles statistiques*. Course notes 3PN1040, Master Physique M2 QLMN, CentraleSupélec and Université Paris-Saclay. 2026.

[2] A. B. Kuper, D. Lazarus, J. R. Manning, and C. T. Tomizuka. "Diffusion in Ordered and Disordered Copper-Zinc". In: *Physical Review* 104.6 (1956), pp. 1536–1541. DOI: 10.1103/PhysRev.104.1536.

[3] C. Kittel. *Introduction to Solid State Physics*. 8th ed. Hoboken, NJ: John Wiley & Sons, 2005.

A Full results of the temperature scan

Table 3: Results of the temperature scan. E: mean energy per atom (eV); C_v: heat capacity of the box (eV/K); χ: susceptibility (eV^{-1}).

T (K)	$\langle E\rangle/N$	C_v	$\langle\lvert\eta\rvert\rangle$	χ	$\langle\alpha_1\rangle$	$\langle\alpha_2\rangle$
200	−2.4840	0.0000	1.000	0	−1.000	1.000
300	−2.4840	0.0002	1.000	0	−1.000	1.000
400	−2.4825	0.0088	0.980	2	−0.962	0.961
500	−2.4788	0.0220	0.926	6	−0.865	0.861
550	−2.4757	0.0277	0.879	10	−0.787	0.779
600	−2.4720	0.0350	0.815	17	−0.692	0.678
650	−2.4673	0.0422	0.717	35	−0.570	0.547
675	−2.4639	0.0523	0.631	86	−0.485	0.455
700	−2.4610	0.0601	0.539	139	−0.409	0.373
710	−2.4596	0.0536	0.487	166	−0.374	0.335
720	−2.4582	0.0602	0.427	211	−0.337	0.295
730	−2.4565	0.0439	0.350	205	−0.296	0.249
740	−2.4556	0.0374	0.313	190	−0.272	0.224
750	−2.4549	0.0322	0.276	183	−0.254	0.205
760	−2.4542	0.0226	0.240	143	−0.235	0.185
775	−2.4535	0.0168	0.212	116	−0.218	0.167
800	−2.4528	0.0128	0.176	94	−0.199	0.146
850	−2.4517	0.0075	0.133	58	−0.170	0.117
900	−2.4510	0.0052	0.108	36	−0.152	0.101
1000	−2.4500	0.0032	0.088	22	−0.126	0.078
1200	−2.4489	0.0017	0.067	11	−0.097	0.054
1600	−2.4478	0.0008	0.056	5	−0.068	0.033

B Code and reproduction

The model is written in Python in `src/model.py` (one module, about 190 lines):

- the array `sous_reseau` holds the species of each site (0 = Cu, 1 = Zn); site i of cell (x, y, z) and sublattice s has index $i = 2(x + 6y + 36z) + s$, so even sites are α and odd sites β;
- `FIRST_NEIGHBOURS` and `SECOND_NEIGHBOURS` are the neighbour tables, built once at import;
- `get_full_energy()` computes Equation (1); `energy_after_flip(E, i, j)` returns the energy after exchanging i and j from the local bonds only, without changing the configuration; `flipping_proba` and `flipping_action` apply the Metropolis rule;
- `cowley()` and `order_parameter()` return (α_1, α_2) and η;
- `metropolis_before_eq(n, T)` returns the energy after each of n steps from B2 (equilibration test); `metropolis(n_eq, n_meas, T, measure_every)` returns the series of E, $|\eta|$, α_1, α_2 measured after equilibration.

The analysis notebook `analysis.ipynb` uses these functions interactively. All the figures and tables of this report are produced by `report/make_figures.py`. From the project folder:

```
python -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python -B report/make_figures.py --rerun   # simulations, figures
cd report && latexmk -pdf report.tex slides.tex       # report and slides
```